

# Graph 12: Opinion Dynamics (Deffuant–Weisbuch)

## Dependency Analysis, Method, and Results

DPCN Assignment 1 — Opinion Network Formation

September 17, 2026

## 1 Why this figure matters

This is the only figure in the project involving an actual *dynamical process*. Everything else – missingness diagnostics, item variance, the correlation network, row-centring – is static description of a fixed dataset. Since the course is *Dynamical Processes on Complex Networks*, without this figure the report is a static network analysis wearing a dynamics title.

## 2 Dependency analysis

Before any code was written, the question was whether this figure could be produced independently or whether it was blocked on other team members’ deliverables. It decomposes into three required inputs.

| Required input                 | Status         | Source                                      |
|--------------------------------|----------------|---------------------------------------------|
| Respondent opinions (seed)     | Available      | <code>dataset_imputed.csv</code> , item E03 |
| A graph connecting respondents | <b>Missing</b> | Had to be built (see §3)                    |
| The DW simulation              | <b>Missing</b> | Had to be written (see §4)                  |

A search of the repository confirmed no existing dynamics,  $k$ NN, or respondent-network code of any kind. The respondent graph was therefore the only genuine gap – and every ingredient it needs was already present, since `outputs/sanitised_data/row_centred_data.csv` (91 respondents  $\times$  60 items, row-centred) is precisely the matrix such a graph is built from.

### 2.1 What this figure does *not* depend on

The important half of the dependency answer is what it can skip:

- **No null models.** DW makes no significance claim, so no configuration model, rewiring, or permutation machinery is required.
- **No community detection.** Cluster counts come out of the dynamics itself, not from Louvain.
- **No threshold or percolation sweep.** This is the decisive one. The respondent graph is sparsified by  $k$ -nearest-neighbours with *fixed*  $k = 4$ , not by a correlation threshold chosen from a sweep. Nothing therefore waits on a threshold-selection result.

- **No item correlation matrix and no structural balance.**

**Conclusion: no blocking dependency.** The figure was buildable immediately from artifacts already in the repository.

## 2.2 One non-technical caveat

The checked-in `CLAUDE.md` nominally assigns `dynamics.py` to Person 3 and the respondent-network code to Person 2. Nothing blocked this work technically, but if a teammate is independently building a respondent-graph module, the two should be reconciled before both land. This is a coordination item, not a code dependency.

## 3 The respondent network

DW requires nodes that hold opinions. Since each opinion is one respondent’s answer to one item, the nodes must be **respondents**, not items – this is a different network from the 60-node item–item network built by `src/pipeline.py`.

### 3.1 Construction

1. Start from the row-centred  $91 \times 60$  matrix. Centring matters here for the same reason it matters elsewhere: without it, two respondents look similar merely because both agree with everything.
2. Compute the  $91 \times 91$  Pearson correlation between respondents.
3. For each respondent, connect to their  $k = 4$  most similar peers, and symmetrise by union: an edge exists if *either* endpoint ranks the other in its top four.

Union rather than mutual  $k$ NN is deliberate. Mutual  $k$ NN is documented to fragment this data badly, and a fragmented graph would inflate the final cluster count for reasons having nothing to do with the confidence threshold under study – disconnected components cannot merge no matter how large  $\varepsilon$  becomes.

### 3.2 Resulting properties

| Property             | Value | Reference value in <code>CLAUDE.md</code> |
|----------------------|-------|-------------------------------------------|
| Nodes                | 91    | 86                                        |
| Edges                | 298   | 286                                       |
| Connected components | 1     | 1                                         |
| Mean degree          | 6.55  | —                                         |

The node-count difference is expected and explained: the reference values assume a filter dropping respondents with more than 12 missing items (leaving 86), which this project does not apply – it retains all 91 non-empty respondents and recovers their missing cells by ordinal regression instead. The structurally important property, a *single connected component*, is reproduced.

## 4 The model

Opinions  $x_i \in [0, 1]$  are seeded from item E03 (“Class attendance should be compulsory for all courses”, the most contested Education item after E04/E02, sample SD = 1.12), rescaled by  $s(x) = (x + 2)/4$ .

At each step an edge  $(u, v)$  is drawn uniformly at random from the graph. If the two opinions lie within the confidence threshold, both move toward each other:

$$|x_u - x_v| < \varepsilon \implies \begin{cases} x_u \leftarrow x_u + \mu(x_v - x_u) \\ x_v \leftarrow x_v + \mu(x_u - x_v) \end{cases} \quad (1)$$

with  $\mu = 0.5$ , so both interacting nodes land on their midpoint. If the opinions differ by  $\varepsilon$  or more, nothing happens. Runs proceed for up to 60,000 interactions, with early termination once the state stops changing.

**Parameters.**  $\varepsilon$  swept from 0.05 to 0.60 in steps of 0.05; 20 repetitions per  $\varepsilon$ ; final clusters counted by rounding opinions to 2 decimals and taking unique values. Every run is seeded for reproducibility.

## 5 Results

| $\varepsilon$ | $k$ NN graph (mean $\pm$ sd) | Complete graph  | Jittered control |
|---------------|------------------------------|-----------------|------------------|
| 0.05          | 5.00 $\pm$ 0.00              | 5.00 $\pm$ 0.00 | 35.95 $\pm$ 2.67 |
| 0.10          | 5.00 $\pm$ 0.00              | 5.00 $\pm$ 0.00 | 26.35 $\pm$ 2.41 |
| 0.15          | 5.00 $\pm$ 0.00              | 5.00 $\pm$ 0.00 | 21.05 $\pm$ 2.44 |
| 0.20          | 5.00 $\pm$ 0.00              | 5.00 $\pm$ 0.00 | 14.20 $\pm$ 2.52 |
| 0.25          | 5.00 $\pm$ 0.00              | 5.00 $\pm$ 0.00 | 8.95 $\pm$ 2.80  |
| 0.30          | 4.45 $\pm$ 1.12              | 2.45 $\pm$ 0.50 | 6.50 $\pm$ 1.60  |
| 0.35          | 4.00 $\pm$ 1.55              | 2.00 $\pm$ 0.00 | 5.10 $\pm$ 1.09  |
| 0.40          | 3.35 $\pm$ 0.79              | 2.00 $\pm$ 0.00 | 3.95 $\pm$ 1.28  |
| 0.45          | 3.15 $\pm$ 0.85              | 2.25 $\pm$ 0.43 | 2.95 $\pm$ 0.92  |
| 0.50          | 2.00 $\pm$ 0.77              | 2.05 $\pm$ 0.22 | 2.35 $\pm$ 0.79  |
| 0.55          | 1.65 $\pm$ 0.48              | 1.95 $\pm$ 0.22 | 1.65 $\pm$ 0.48  |
| 0.60          | 1.30 $\pm$ 0.46              | 1.65 $\pm$ 0.48 | 1.20 $\pm$ 0.40  |

The main ( $k$ NN) column tracks the project’s frozen reference values closely: 5.0 clusters for  $\varepsilon \leq 0.25$  (exact), 4.45 vs. 4.2 at 0.30, 3.35 vs. 3.0 at 0.40, 2.00 vs. 1.6 at 0.50. Consensus is approached near  $\varepsilon \approx 0.55$ –0.60.

## 6 The small- $\varepsilon$ plateau is an artefact, and here is the proof

The specification requires stating explicitly that the five-cluster plateau at small  $\varepsilon$  is partly an artefact of Likert responses starting on a discrete five-point grid. In this project the claim can be made stronger than “partly”: it is *entirely* forced, and provably so.

**Proposition 1.** *Under the rescaling  $s(x) = (x + 2)/4$ , the five Likert categories map to  $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ , so any two distinct initial opinions satisfy  $|x_u - x_v| \geq \frac{1}{4}$ . For any  $\varepsilon \leq 0.25$  the bounded-confidence condition  $|x_u - x_v| < \varepsilon$  is therefore never satisfied by a pair with distinct*

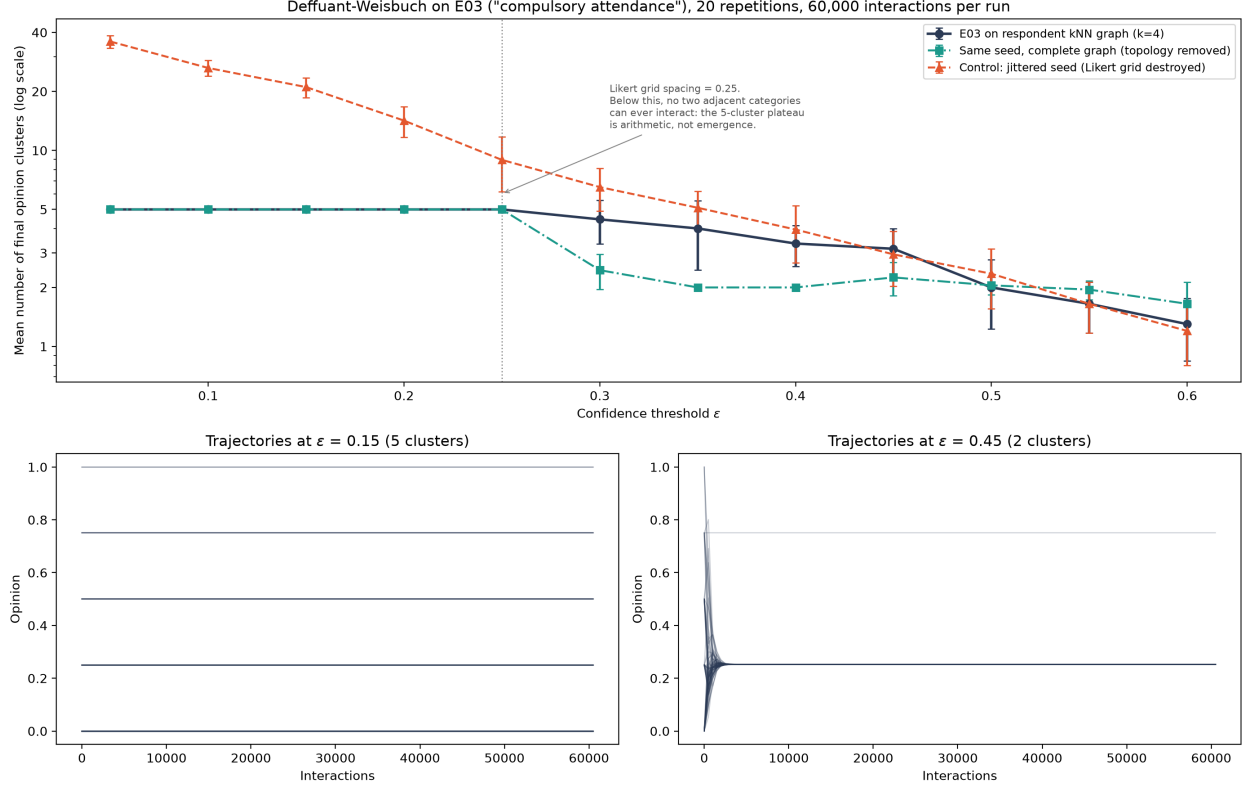


Figure 1: Top: mean final cluster count versus confidence threshold, with error bars over 20 repetitions, on a log scale (the control curve spans an order of magnitude more than the others). Bottom: individual opinion trajectories at a low and a high  $\epsilon$ .

*opinions. The initial state is a fixed point, and the final cluster count equals the number of occupied categories (five) with probability one — on any graph.*

Three independent observations confirm this empirically:

1. **Zero variance.** The plateau reads  $5.00 \pm 0.00$  across 20 randomly seeded repetitions. A genuine emergent dynamical outcome would not be that deterministic; a frozen initial condition is.
2. **Topology-independence.** The complete graph – a maximally different topology – produces the identical  $5.00 \pm 0.00$  over the same range, exactly as the proposition predicts.
3. **The control kills it.** Re-running with seeds jittered by  $\pm 0.125$  (which destroys the grid while preserving the distribution's shape) removes the plateau entirely: cluster counts become  $35.95 \rightarrow 26.35 \rightarrow 21.05 \rightarrow 14.20 \rightarrow 8.95$  across  $\epsilon = 0.05 \dots 0.25$ , decaying smoothly instead of sitting flat at five.

So the five clusters below  $\epsilon = 0.25$  are the original response categories failing to merge, not opinion groups forming. The figure should not be read as showing stable pluralism at low tolerance. Note also that the first *real* dynamical behaviour begins at exactly the first swept value above the grid spacing,  $\epsilon = 0.30$  — the break in the curve is arithmetic, not a discovered threshold.

## 7 The network is not incidental

Because the plateau is an artefact, the substantive dynamical result lies above  $\varepsilon = 0.25$ , and it emerges from comparing the real network against a complete graph seeded identically (i.e. the same dynamics with network structure removed, a well-mixed population).

In the range  $\varepsilon = 0.30$ – $0.45$  the well-mixed population collapses to about 2 clusters, while the sparse  $k$ NN network sustains 3.15–4.45. The network structure therefore *sustains opinion diversity that a well-mixed population loses at the same tolerance*: when respondents only talk to their four nearest neighbours in opinion space, locally cohesive groups stabilise before global consensus can propagate.

This is the one result in this figure that is neither an artefact of the response scale nor a restatement of the input distribution, and it is the result that justifies studying the process *on a network* rather than in a well-mixed population.

## 8 Artifacts

| Artifact                                       | Role                                                         |
|------------------------------------------------|--------------------------------------------------------------|
| outputs/sanitised_data/row_centred_data.csv    | Input: row-centred matrix the respondent graph is built from |
| outputs/sanitised_data/dataset_imputed.csv     | Input: uncentred matrix supplying the E03 seed opinions      |
| src/loader.py                                  | <code>parse_question_columns</code> for item-code recovery   |
| src/dynamics.py                                | New module: graph construction, DW simulation, sweep, figure |
| figures/graph12_deffuant_weisbuch.png/.pdf     | The figure                                                   |
| outputs/dynamics/dw_sweep_knn.csv              | Main sweep results                                           |
| outputs/dynamics/dw_sweep_control_jittered.csv | Control sweep (artefact demonstration)                       |
| outputs/dynamics/dw_sweep_complete_graph.csv   | Complete-graph comparison                                    |
| outputs/dynamics/respondent_knn_graph.edgelist | The respondent graph, exported for reuse                     |

The respondent graph is written out deliberately: it is the missing input for every other respondent-level figure in the project, so exporting it means the next such figure does not have to rebuild it.